RELATIVE PSEUDO-COHERENCE IS UNCHANGED BY POST-COMPOSING WITH A FLAT FINITELY PRESENTED MIDDLE MAP

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 20682-20689 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 20691-20697, the complete author proof (5 body lines).

Editorial note. Navigation stubs: 1 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Second pillar (with L16115) of the fppf-base descent card L16096 collected in wave 12; the citation chain closes.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Chapter lemma composition-relative-pseudo-coherent (card L16115); existence of affine-space factorizations for locally finite type maps; flat finitely presented stability of affine spaces.

Source reference. Original lemma 09UT: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 17235–17245. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $X \to T \to S$ be morphisms of schemes. Assume $T \to S$ is flat and locally of finite presentation and $X \to T$ locally of finite type. Let $E \in D(\mathcal{O}_X)$. Let $m \in \mathbf{Z}$. Then E is m-pseudo-coherent relative to S if and only if E is m-pseudo-coherent relative to T.*

Proof. Locally on X we can choose a closed immersion $i : X \to \mathbf{A}^n_T$. Then $\mathbf{A}^n_T \to S$ is flat and locally of finite presentation. Thus we may apply Lemma 09UT to see the equivalence holds. □

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.